Earning & Managing Money
Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Last week, Paola earned $358.40 when she worked 16 hours at a normal rate in her part-time job. This week she worked 12 hours at a normal rate and 3 hours at time-and-a-half. How much did she earn this week? 1

normal rate $= \$358.40 \div 16 = \$22.40$

earnings $= 12 \times 22.40 + 3 \times 1.5 \times 22.40 = \$369.60$

2 Alessia works 42 hours per week at a rate of $32.50 per hour. She decides to take four weeks of annual leave. While she is on annual leave, she is paid a loading of 17.5% on her wage. What is Alessia's total wage for the period of time that she is on annual leave? 1

$42 \times 32.50 \times 4 \times 117.5\%$

$6415.50

3 During the previous financial year, Diego earned a gross salary of $79 348 for his work as a forensic statistician. Diego also earned $8 920 from being a Twitch streamer. All up, Diego's taxable income has been calculated to be $83 356.

a. Calculate how much Diego is claiming in tax deductions for his streaming/work equipment and home energy consumption. 1

$79348 + 8920 - 83356 = \$4912$

b. Using the tax table below, calculate Diego's tax payable to the nearest cent. 2

Taxable income	*Tax payable on this income*
0 – $18 200	Nil
$18 201 – $45 000	19 cents for each $1 over $18 200
$45 001 – $120 000	$5092 plus 32.5 cents for each $1 over $45 000
$120 001 – $180 000	$29 467 plus 37 cents for each $1 over $120 000
$180 001 and over	$51 667 plus 45 cents for each $1 over $180 000

$5092 + 0.325(83356 - 45000) = \$17\,557.70$

c. Diego's employer has already taken $11 905 as PAYG tax. Calculate Diego's refund or liability, justifying your answer with calculations to the nearest cent. Do not include the Medicare Levy in your calculation. $17\,557.70 - 11\,905 = 5652.70$ 1

4 A train trip costs \$ 6.80 including 10% GST. What is the train trip's price before GST, correct to the nearest cent? 1

\$6.18

5 The table shows the income tax rates for the 2020-2021 financial year. 3

Taxable income	*Tax payable on this income*
0 – \$18 200	Nil
\$18 201 – \$45 000	19 cents for each \$1 over \$18 200
\$45 001 – \$120 000	\$5092 plus 32.5 cents for each \$1 over \$45 000
\$120 001 – \$180 000	\$29 467 plus 37 cents for each \$1 over \$120 000
\$180 001 and over	\$51 667 plus 45 cents for each \$1 over \$180 000

Wendy has a gross annual salary of \$94 000. She has allowable tax deductions of \$1900 for home-office equipment and \$470 for union fees. Wendy must also pay a Medicare Levy of 2% of her taxable income. Calculate the total tax payable by Wendy including the Medicare Levy.

$$\text{Taxable income} = 94\,000 - 1900 - 470 = 91\,630$$
$$\text{Income Tax} = 5092 + 0.325 \times (91630 - 45000) = 20246.75$$
$$\text{Medicare Levy} = 0.02 \times 91630 = 1832.60$$
$$\text{Total Tax Payable} = 20246.75 + 1832.60 = 22079.35$$

6 Jonas is a real estate agent. He earns \$600 per week plus commission on any sales he makes. His commission is calculated using the schedule below. Last week, Jonas sold a block of land for \$ 900,000. Calculate Jonas' income for that week. 2

Value of Sale	Commission
Less than \$260,000	4%
\$260,001 - \$ 520,000	\$10,400 plus 2% of each dollar over \$260,000
Over \$520,000	\$15,600 plus 1.5% of each dollar over \$520,000

15600 + 1.5% (900000 − 520000) = 21300
Income = 600 + 21300 = 21900

7 This table is used to calculate income tax for the 2021 – 2022 financial year. 4

Taxable income	*Tax on this income*
0 – \$18,200	Nil
\$18,201 – \$45,000	19 cents for each \$1 over \$18,200
\$45,001 – \$120,000	\$5092 plus 32.5 cents for each \$1 over \$45,000
\$120,001 – \$180,000	\$29,467 plus 37 cents for each \$1 over \$120,000
\$180,000 and over	\$51,667 plus 45 cents for each \$1 over \$180,000

Max is a lawyer and has an annual gross salary of \$116,487. He has allowable tax deductions of \$980 for renewing his practising certificate and \$650 of travel expenses between clients annually. Max must also pay a Medicare Levy of 2% of his taxable income. During the year, he paid \$3000 per month in Pay As You Go (PAYG) tax. Will Max receive a tax refund or have to pay additional tax? Justify your answer with calculations to determine the amount of tax refund or tax payable.

Taxable income = 116487 − 980 − 650 = \$114,857
Income tax = (114857 − 45000) × 0.325 + 5092 = \$27,795.53
Medicare = 114857 × 0.02 = \$2297.14
Total Tax = 2297.14 + 27795.53 = \$30,092.67
So Max's refund = 3000 × 12 − 30092.67 = \$5907.33

8 Stephanie's cookbook sells for \$29.00 per copy. She earns a royalty of 12% of the selling price of the book. How much money will Stephanie receive if 7200 books are sold in total? 1

$(29)(0.12)(7200)$ \$25 056

9 When Adam joined a sales company he was offered a choice of two different salary contracts. The details of the contracts are outlined in the table below.

	Salary	End of year commission on total sales
Contract A	\$65 000	2%
Contract B	\$60 000	3%

a. Find how much Adam would earn under each contract in a year if his total sales were \$400 000. 1

Contract A = \$65 000 + (\$400 000 × 2%) = \$73 000
Contract B = \$60 000 + (\$400 000 × 3%) = \$72 000

b. Another employee, Lottie, earned \$90 000 in a particular year. She is on Contract A. Find her total sales for that year. 2

Contract A:

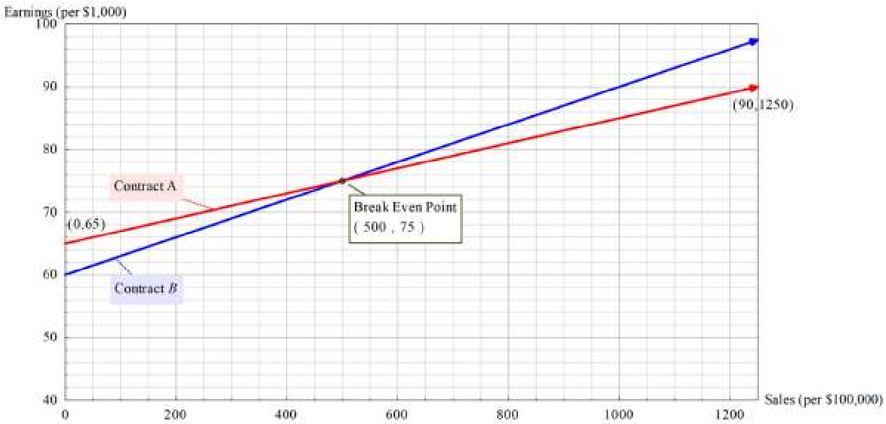
$$\$90\,000 = \$65\,000 + (\text{sales} \times 2\%)$$
$$\$90\,000 - \$65\,000 = \text{sales} \times 2\%$$
$$\frac{\$90\,000 - \$65\,000}{0.02} = \text{sales}$$

Sales = \$1 250 000

c. Write a linear equation to model earnings, E, in relation to sales, S, for Contract A. 1

Contract A: $E = 65\,000 + 0.02\,S$

d. On the grid below, draw the graph of the linear relationship for Contract A and label it. 1



e. From your graph or otherwise, determine the total sales for which Contract A and Contract B earn the same amount of money. 1

Sales \$500 000 have equal earnings of \$75 000 under each contract.

f. Recommend which contract Adam and Lottie should be on, based on your calculations above. 2

Adam should chose Contract A. As a starting salesperson he will benefit from having the larger retainer.
Lottie should move to Contract B. She will earn more with a larger commission from her sales.

10 A video game console costs \$389, inclusive of 10% goods and services tax (GST). What is the price of the video game excluding GST, correct to the nearest dollar? 1

Using the unitary method gives:

$$\frac{389}{1.1} = 353.6363\ldots \approx \$354$$

11 The table shows the income tax rates for the 2020–2021 financial year.

Taxable income	*Tax on this income*
0–\$18 200	Nil
\$18 201–\$45 000	19c for each \$1 over \$18 200
\$45 001–\$120 000	\$5092 plus 32.5c for each \$1 over \$45 000
\$120 001–\$180 000	\$29 467 plus 37c for each \$1 over \$120 000
\$180 001 and over	\$51 667 plus 45c for each \$1 over \$180 000

For the 2020–2021 financial year, Hara had a total income of \$145 000, and her tax payable was \$33 167 (excluding the Medicare levy).

a. Calculate Hara's taxable income. 2

Let y be Hara's taxable income. Using the bracket \$120 001–\$180 000 gives:

$$29467 + 0.37(y - 120000) = 33167$$
$$0.37(y - 120000) = 33167 - 29467$$
$$y - 120000 = \frac{3700}{0.37}$$
$$y = 10000 + 120000 = \$130000$$

b. How much did Hara claim as allowable tax deductions? 1

$$\text{allowable tax deduction} = \text{income} - \text{taxable income}$$
$$= 145000 - 130000 = \$15000$$

12 Jim, a landscaping contractor, charges by the hour for his company's services. To complete a particular job, he will have to use three workers and pay each of them \$20 per hour. The fixed costs for the job are \$150 and it will take four hours to complete the job. To break even on this job, what should his hourly charge to the client be? 1

$$\text{hourly rate} = \frac{3 \times 20 \times 4 + 150}{4}$$
$$= \$97.5/\text{hour.}$$

Earning & Managing Money
Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 Lauren and Andy work in the same restaurant, they each earned \$576 last week. Andy worked 4 more hours than Lauren, and was paid time-and-a-half for these extra hours. If Lauren is paid \$16/hour, find Andy's hourly pay rate? 1

\$13.71

2 Malik's normal rate of pay is \$19.75 per hour. In one week he worked 17 hours at the normal rate, 7 hours at time-and-a-half and 2 hours double time. He was also paid a wet weather allowance of \$65 for the week. What were his total earnings for the week? 2

$(19.75 \times 17) + (19.75 \times 1.5 \times 7) + (19.75 \times 2 \times 2) + 65$

$= \$687.125 = \$687.13$

3 Two painters each provide a quote for painting an area of 1500 square metres. 3

Painter A: charges \$90 per 25 square metres.

Painter B: charges \$80 per hour and bases his quote on painting 20 square metres per hour.

How much will be saved by choosing the cheaper quote? Justify your answer using mathematical reasoning and /or calculations.

Cost painter A $= \frac{1500}{25} = 60$

$= 60 \times 90 = \$5400$

Cost painter B $= \frac{1500}{20} = 75$

$= 75 \times 80 = \$6000$

Saving $= \$600$

4 Toby is a real estate agent. He earns \$950 per month plus commission on any sales he makes. His commission is calculated using the schedule below. Last month, Toby sold a block of land for \$180 000, a house for \$650 000 and a town house for \$490 000. Calculate Toby's income for that month. 3

Value of Sale	Commission
Less than \$260 000	4%
\$260 001 – \$520 000	\$10 400 plus 2% of each dollar over \$260 000
Over \$520 000	\$15 600 plus 1.5% of each dollar over \$520 000

Total value of sales =\$180 000 + 650 000 + 490 000 - \$1 320 000

Commission = 15 600 + 1.5% of (1 320 000 – 520 000) = \$27 600

Income for the month = \$27 600 + 950 = \$28 550

5 The table shows the income tax rates for the 2021-2022 financial year.

Michelle earned a gross annual salary of $92 500. She has allowable tax deductions of $1200 for home office expenses. Michelle must also pay a Medicare Levy of 2% of her taxable income.

Taxable Income	Tax payable on this income
0 – $18,200	Nil
$18,201 – $45,000	19 cents for each $1 over $18,200
$45,001 – $120,000	$5,092 plus 32.5 cents for each dollar over $45,000
$120,001 – $180,000	$29,467 plus 37 cents for each $1 over $120,000
$180,001 and over	$51,667 plus 45 cents for each $1 over $180,000

a. Calculate the amount of tax payable by Michelle including the Medicare Levy. 3

Taxable income = gross income – deductions
Taxable income = 92 500 – 1200 = $91 300
Tax bracket from the table:

Taxable Income	*Tax payable on this income*
0 – $18,200	Nil
$18,201 – $45,000	19 cents for each $1 over $18,200
$45,001 – $120,000	$5,092 plus 32.5 cents for each dollar over $45,000
$120,001 – $180,000	$29,467 plus 37 cents for each $1 over $120,000
$180,001 and over	$51,667 plus 45 cents for each $1 over $180,000

Tax payable = $5 092 + 0.325 × (91 300 – 45 000) = $20 139.50
Medicare Levy = 2% of taxable income = 0.02 × 91 300 = $1 826
Total payable = $20 139.50 + $1 826 = $21 965.50

b. Michelle's employer has deducted $1850 per month from Michelle's pay in Pay As You Go (PAYG) tax. Determine if Michelle will receive a tax refund or will have to pay a tax bill. Show calculations, including the amount of the refund or bill. 1

Total $PAYG$ paid = 12 × $1850 = $22 200
This is more than the $21 965.50 Michelle needed to pay so she will receive a refund.
Refund = $22 200 – 21 965.50 = $234.50

6 Stephanie works part time in a medical centre part and receives a yearly salary of $34479. She also receives an income of $1950 per year from babysitting. Her total deductions are $1570. During the year she paid tax instalments of $3950.

Taxable income	*Tax on this income*
0–$18 200	Nil
$18 201–$37 000	19c for each $1 over $18 200
$37 001–$90 000	$3572 plus 32.5c for each $1 over $37 000
$90 001–$180 000	$20 797 plus 37c for each $1 over $90 000
$180 001 and over	$54 097 plus 45c for each $1 over $180 000

a. Find Stephanie's taxable income. 1

34479 + 1950 – 1570 = 34859

b. Using the tax table given below, find the income tax payable. 2

0.19 (34859 – 18200) = 3165.21

c. Calculate the Medicare levy that Stephanie needs to pay. 697.18 1

d. Find Stephanie's tax refund. 2

3165.21 + 697.18 = 3862.39
Refund = 3950 – 3862.39 = $ 87.61

7 Natalie has a gross income of \$1585 a fortnight, every fortnight of the year.

Natalie receives dividends of \$106 throughout the year, spends \$180 on work related expenses and \$989 on a new mobile phone for personal use. She also receives 1.2% simple interest on a savings account with a balance of \$212 045.

Natalie also donates \$50 to charity each month and deducts this from her taxes.

a. Show that Natalie's taxable income is approximately \$43 081. 3

$$\begin{aligned} \textit{Gross income} &= (1\,585 \times 26)_{\text{Salary}} + 106_{\text{dividends}} \\ &\quad + 0.012 \times 212045_{\text{interest}} \\ &= \$43\,860.54 \approx \$43\,861 \end{aligned}$$

$$\text{Deductions} = 180_{\text{work expenses}} + 50 \times 12_{\text{donations}} = \$780$$

$$\text{Taxable income} = \$43\,081$$

b. The table below shows different tax payable in Australia according to taxable income. 3

Taxable income (\$)	*Tax payable*
\$0 – \$18 200	Nil
\$18 201 – \$37 000	19 cents for each \$1 over \$18 200
\$37 001 – \$90 000	\$3 572 plus 32.5 cents for each \$1 over \$37 000
\$90 001 – \$180 000	\$20 797 plus 37 cents for each \$1 over \$80 000
Over \$180 001	\$54 096 plus 45 cents for each \$1 over \$180 000

The Medicare levy in Australia is calculated at 2% of taxable income.

If Natalie has already paid \$6887 in tax, will she receive a tax refund or owe money?

Justify your answer with calculations.

$$(3\,572 + 0.325(43\,081 - 37\,000))_{\textit{Based on tax bracket}}$$
$$+ 0.02 \times 43\,081_{\textit{medicare levy}}$$
$$6887 - 6409.95 = 477.05$$

Natalie will receive a tax return of \$477.05

8 The table below shows the fortnightly Abstudy allowances available to independent indigenous students. Cory and Emma are indigenous students in a de facto relationship with no children. Cory is 24 while Emma is 21. Calculate their combined fortnightly allowance. 2

Fortnightly Abstudy Allowances

Conditions for Eligibility	Maximum Fortnightly Payment
Single, no children, living at home	
Under 16	\$34.80
16–17 years	\$253.30
18–21 years	\$304.60
22 years or older	\$565.70
Single, no children, living away from home	
Under 16	\$462.50
16–21 years	\$462.50
22 years or older	\$565.70
Partnered, no children	
16–21 years	\$462.50
22 years or older	\$510.80
Single, with dependent child	
16–21 years	\$606.00
22 years or older	\$612.00
Partnered, with dependent child	
16–21 years	\$507.90
22 years or older	\$510.80

Cory's allowance = \$510.80

Emma's allowance = \$462.50

$$\begin{aligned}\text{Combined allowance} &= 510.80 + 462.50 \\ &= \$973.30\end{aligned}$$

9 Henry is paid an hourly rate and receives penalty rates for weekends. His total pay for one particular week is shown in the pay slip below. His hourly rate is missing in the pay slip. 3

Calculate his hourly rate.

Employee	Henry		
Date	1/7/20-7/7/20		
Normal hours	15		
Time-and-a-half	6	Hourly rate:	Total pay:
Double time	4		\$561.92

Number of hours paid = $15 + (1.5 \times 6) + (2 \times 4) = 32$ hours

Hourly Rate = $\$561.92 \div 32 = \$17.56$